

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3,BL4)

Assessment Tool Weightage: 40%

APPLICATION-BASED QUESTIONS

Code of Conduct:

*I **Kartheeka Repalle (192372289)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kartheeka Repalle

APPLICATION-BASED QUESTIONS WITH DETAILED ANSWERS

Question 1

Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Answer:

The classical encryption technique used is the Caesar Cipher with a shift value of +4. Each letter in the plaintext is replaced by the letter four positions ahead in the English alphabet.

Plaintext : MEET AT NOON

Shift : +4

Ciphertext: QIIX EX RSSR

Step-by-step Encryption:

M→Q

E→I

E→I

T→X

$A \rightarrow E$

$T \rightarrow X$

$N \rightarrow R$

$O \rightarrow S$

$O \rightarrow S$

$N \rightarrow R.$

Decryption:

The receiver shifts every ciphertext letter four positions backward.

$Q \rightarrow M$

$I \rightarrow E$

$I \rightarrow E$

$X \rightarrow T$

$E \rightarrow A$

$X \rightarrow T$

$R \rightarrow N$

$S \rightarrow O$

$S \rightarrow O$

$R \rightarrow N.$

Recovered Message:

MEET AT NOON

Analysis:

This method ensures that only users who know the shift value can recover the original message. Although simple and fast, Caesar Cipher provides weak security because attackers can easily try all possible shifts.

Conclusion:

The encrypted message is QIIX EX RSSR, and applying the reverse shift restores the original message.

Question 2

Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Deduce the most likely original message. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Answer:

The intercepted message is encrypted using the Atbash Cipher, where each alphabet letter is replaced by its opposite ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$...).

Ciphertext : GSRH RH Z HVXIVG

Plaintext : THIS IS A SECRET

Justification:

- The repeated word 'RH' decrypts to 'IS', a common English word.
- The one-letter word 'Z' decrypts to 'A'.
- The reverse alphabet mapping correctly converts every letter.
- The final output forms a meaningful English sentence.

Analysis:

Atbash is a monoalphabetic substitution cipher with a fixed substitution pattern. Since no secret key is required, it is easy to decrypt if the cipher type is known.

Conclusion:

The original message is THIS IS A SECRET.

Question 3

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

Demonstrate how the encryption process transforms the message step-by-step. Explain how the repeating keyword influences each character of the ciphertext.

Answer:

The Vigenère Cipher is a keyword-based encryption method.

Plaintext : H E L L O

Keyword : K E Y K E

Encryption:

$H + K = R$

$E + E = I$

$L + Y = J$

$L + K = V$

$O + E = S$

Ciphertext: RIJVS

Explanation:

The keyword is repeated until it matches the message length. Each keyword letter determines the shift applied to the corresponding plaintext character, making the encryption stronger than a fixed Caesar shift.

Conclusion:

The encrypted word is RIJVS.

Question 4

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection. Analyze how investigators could detect the presence of hidden data within the image. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Answer:

Hidden information inside an image can be detected using steganalysis.

Detection Methods:

1. Analyze image pixels for unusual patterns or statistical changes.
2. Examine image metadata and compare the suspicious image with the original.
3. Use forensic tools to identify hidden payloads or modified least significant bits (LSBs).

Security Enhancements:

- Encrypt the secret message before embedding it.
- Use key-based or randomized steganography so only authorized users can extract the data.

Analysis:

Combining encryption with steganography protects both the content and the existence of the secret message.

Conclusion:

Steganalysis helps detect hidden data, while encryption and key-based embedding improve security.

Question 5

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as:

“WECRLTEERDSOEFEAOCAIVDEN”.

a. Determine the most probable original message by reversing the encryption process. b. Critically evaluate why such a method is vulnerable in modern communication systems. c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Answer:

This is a Rail Fence Cipher with three rails.

a) Original Message:

WE ARE DISCOVERED FLEE AT ONCE

b) Vulnerabilities:

- Preserves letter frequency.
- Easy to break using brute-force and pattern analysis.
- Does not provide authentication or integrity.
- Not suitable for modern secure communication.

c) Practical Enhancement:

Use a modern encryption algorithm such as AES-256 or combine transposition with substitution and secure key management.

Analysis:

Modern encryption algorithms provide confidentiality, integrity, and strong resistance against cryptographic attacks.

Conclusion:

The Rail Fence Cipher is useful for learning cryptography but should not be used to protect sensitive information today.